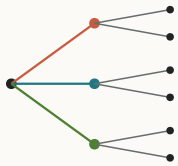
# Permutations and combinations CHEAT SHEET



## COUNTING PRINCIPLES



3 × 2 = 6 leaves

every first choice goes with every second choice

and: m ways, then n ways → $m \times n$

3 T-shirts × 2 trousers = 6 outfits

café: $3 \times 4 \times 2 = 24$ lunches

4-digit PIN: $10 \times 10 \times 10 \times 10 = 10\,000$

or (no overlap): $m + n$  5 pizzas or 3 burgers: 8

overlap: 12 football + 9 basketball − 5 both = 16

## RESTRICTIONS

fill the hardest place first

3-digit numbers: $9 \times 10 \times 10 = 900$

all digits different: $9 \times 9 \times 8 = 648$

odd, all different: $5 \times 8 \times 8 = 320$ (last, first, middle)

code: 2 different letters, 3 digits: $26 \times 25 \times 1000 = 650\,000$

count the opposite: $10^4 - {}^{10}P_4 = 4960$ PINs with a repeat

## FACTORIALS

$n! = n \times (n-1) \times \ldots \times 2 \times 1$

$n! = n \times (n-1)!$  $0! = 1$

$\frac{8!}{6!} = 8 \times 7 = 56$  $\frac{10!}{8!\,2!} = 45$

n things in a row: n! orders  $52! \approx 8.07 \times 10^{67}$

| 0! | 1! | 2! | 3! | 4! | 5! | 6! | 7! | 8! |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | 6 | 24 | 120 | 720 | 5040 | 40320 |

## PERMUTATIONS

${}^nP_r = n(n-1)\ldots(n-r+1) = \frac{n!}{(n-r)!}$

${}^8P_3 = 8 \times 7 \times 6 = 336$ (gold, silver, bronze)

${}^{10}P_4 = 10 \times 9 \times 8 \times 7 = 5040$ PINs, no repeats

${}^nP_n = n!$  ${}^nP_1 = n$  ${}^nP_0 = 1$

repeats allowed: $n^r$  $10^4 = 10\,000$, $5^3 = 125$

${}^nP_2 = 56 \Rightarrow n(n-1) = 56 \Rightarrow n = 8$

## IDENTICAL OBJECTS

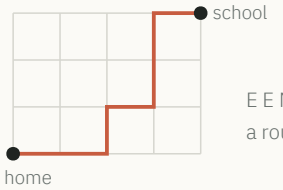
n things, p alike, q alike, …: $\frac{n!}{p!\,q!\,\ldots}$

BOOK: $\frac{4!}{2!} = 12$  BANANA: $\frac{6!}{3!\,2!} = 60$

MISSISSIPPI: $\frac{11!}{4!\,4!\,2!} = 34\,650$

COFFEE: $\frac{6!}{2!\,2!} = 180$

grid walk, 4 east, 3 north: $\frac{7!}{4!\,3!} = 35$ routes




E E N E N N E: one route

a route = an order of 4 E's, 3 N's

## CONDITIONS

together: glue a block  $4! \times 2! = 48$ (5 in a row)

apart: all − together  $120 - 48 = 72$

round a table: $(n-1)!$  5 people: $4! = 24$

PLANET, starting with a vowel: $2 \times 5! = 240$

ARRANGE, the R's apart: $1260 - 360 = 900$

turning the table changes nothing:

5! rows ÷ 5 turns = 4!

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## COMBINATIONS

${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$ (r numbers counting down from n, over r!)

${}^{n}P_{r} = {}^{n}C_{r} \times r!$ choose, then arrange

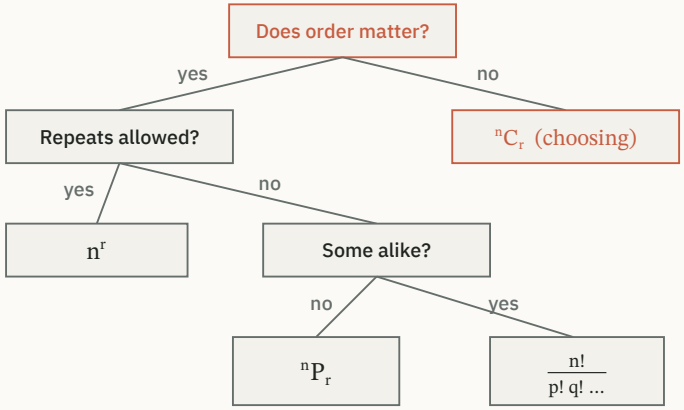
${}^{10}C_{3} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$

2 toppings of 4: ${}^{4}C_{2} = 6$ handshakes, 10 people: ${}^{10}C_{2} = 45$

5-card hands: ${}^{52}C_{5} = 2\,598\,960$

${}^{n}C_{0} = {}^{n}C_{n} = 1$ ${}^{n}C_{1} = n$

## WHICH FORMULA?




test: swap two things in an answer.
A different answer → order matters.

cup of 3 scoops from 10: ${}^{10}C_{3} = 120$ cone: ${}^{10}P_{3} = 720$

committee of 3 from 10: 120 3 named roles: 720

## PROPERTIES

${}^{n}C_{r} = {}^{n}C_{n-r}$ ${}^{10}C_{2} = {}^{10}C_{8} = 45$

${}^{n}C_{x} = {}^{n}C_{y} \Rightarrow x = y$ or $x + y = n$

${}^{20}C_{18} = {}^{20}C_{2} = 190$ ${}^{n}C_{2} = 28 \Rightarrow n = 8$

Pascal's rule: ${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$

subsets of n things: $2^{n}$ 4 toppings: $16 = 1 + 4 + 6 + 4 + 1$

## PASCAL'S TRIANGLE

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | | | | | = 1 |
| | | 1 | | 1 | | | | = 2 |
| | 1 | | 2 | | 1 | | | = 4 |
| 1 | | 3 | | 3 | | 1 | | = 8 |
| | 1 | | 4 | | 6 | | 4 | 1 = 16 |
| 1 | | 5 | | 10 | | 10 | | 5 1 = 32 |
| | 1 | | 6 | | 15 | | 20 | 15 6 1 = 64 |
| 1 | | 7 | | 21 | | 35 | | 35 21 7 1 = 128 |

row n: ${}^{n}C_{0}, {}^{n}C_{1}, \ldots, {}^{n}C_{n}$ each = the two above: $3 + 3 = 6$

## MIXED PROBLEMS

choose, then arrange, then multiply

DAUGHTER, 3 vowels + 2 consonants: ${}^{3}C_{3} \times {}^{5}C_{2} \times 5! = 1200$

4 of 9, then a captain: ${}^{9}C_{4} \times 4 = 9 \times {}^{8}C_{3} = 504$

exactly 2 girls (6 girls, 5 boys, team of 4): ${}^{6}C_{2} \times {}^{5}C_{2} = 150$

at least one boy = all − none: ${}^{11}C_{4} - {}^{6}C_{4} = 315$

10 points on a circle: ${}^{10}C_{2} = 45$ lines, ${}^{10}C_{3} = 120$ triangles

diagonals of an n-gon: ${}^{n}C_{2} - n$ octagon: 20

1 goalkeeper of 2, 10 of 13: ${}^{2}C_{1} \times {}^{13}C_{10} = 572$

6 round a table, 2 together: $4! \times 2 = 48$

4 of 7 in a row of seats: ${}^{7}P_{4} = {}^{7}C_{4} \times 4! = 840$

## ONE KEYPAD, THREE ANSWERS

| | | |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |
| * | 0 | # |

any 4 digits: $10^{4} = 10\,000$

no repeats, in order: ${}^{10}P_{4} = 5040$

a set of 4 digits: ${}^{10}C_{4} = 210$